LISTA de G.A. Gabriela Vinha Feitosa

1- a) A+2B

$$\begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix} + 2 \cdot \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix} + \begin{pmatrix} 0 & 10 \\ 6 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 10 \\ 2 & 3 \end{pmatrix}$$

b) AB - BA

$$\begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix} \cdot \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} - \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix}$$

$$\begin{pmatrix} 1\cdot0+0\cdot3 & 1\cdot5+0\cdot(-2) \\ 2\cdot0+7\cdot3 & 2\cdot5+7\cdot(-2) \end{pmatrix} - \begin{pmatrix} 0\cdot1+5\cdot2 & 0\cdot0+5\cdot7 \\ 3\cdot1+(-2)\cdot2 & 3\cdot0+(-2)\cdot7 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 5 \\ 21 & -4 \end{pmatrix} - \begin{pmatrix} 10 & 35 \\ -1 & -14 \end{pmatrix}$$

$$\therefore AB-BA = \begin{pmatrix} -10 & -30 \\ 22 & 10 \end{pmatrix}$$

c) 2C - D

$$2 \cdot \begin{pmatrix} -2 & 3 & -7 \\ 7 & -3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} -4 & 6 & -14 \\ 14 & -6 & -4 \end{pmatrix} \cdot \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} =$$

$$= \begin{pmatrix} -4\cdot(-3)+6\cdot1+(-14)\cdot(-2) & -4\cdot2+6\cdot1+(-14)\cdot0 & -4\cdot0+6\cdot4+(-14)\cdot2 \\ 14\cdot(-3)+(-6)\cdot1+(-4)\cdot(-2) & 14\cdot2+(-6)\cdot1+(-4)\cdot0 & 14\cdot0+(-6)\cdot4+(-4)\cdot2 \end{pmatrix} = \begin{pmatrix} 46 & -2 & -4 \\ -40 & 22 & -32 \end{pmatrix}$$

d) $2D^t - 3E^t$

$$2 \cdot \begin{pmatrix} -3 & 1 & -2 \\ 2 & 1 & 0 \\ 0 & 4 & 2 \end{pmatrix} - 3 \cdot \begin{pmatrix} 2 & -1 & -6 \\ 4 & 0 & 0 \\ -3 & -4 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -6 & 2 & -4 \\ 4 & 2 & 0 \\ 0 & 8 & 4 \end{pmatrix} - \begin{pmatrix} 6 & -3 & -18 \\ 12 & 0 & 0 \\ -9 & -12 & -3 \end{pmatrix}$$

$$\therefore 2D^t - 3E^t \begin{pmatrix} -12 & 5 & 14 \\ -8 & 2 & 0 \\ 9 & 20 & 7 \end{pmatrix}$$

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e) $D^2 + DE$

$$\begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix}^2 + \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} -3 & 2 & 0 \\ 1 & 1 & 4 \\ -2 & 0 & 2 \end{pmatrix} + \begin{pmatrix} (-3)\cdot 2 + 2\cdot(-1) + 0\cdot(-6) & (-3)\cdot 4 + 2\cdot 0 + 0\cdot 0 & -3\cdot(-3) + 2\cdot(-4)\cdot 0\cdot(-1) \\ 1\cdot 2 + 1\cdot(-1) + 4\cdot(-6) & 1\cdot 4 + 1\cdot 0 + 4\cdot 0 & 1\cdot(-3) + 1\cdot(-4) + 4\cdot(-1) \\ (-2)\cdot 2 + 0\cdot(-1) + 2\cdot(-6) & -2\cdot 4 + 0\cdot 0 + 2\cdot 0 & -2\cdot(-3) + 0\cdot(-4) + 2\cdot(-1) \end{pmatrix}$$

$$\begin{pmatrix} (-3)\cdot(-3) + 2\cdot 1 + 0\cdot(-2) & (-3)\cdot 2 + 2\cdot 1 + 0\cdot 0 & (-3)\cdot 0 + 2\cdot 4 + 0\cdot 2 \\ 1\cdot(-3) + 1\cdot 1 + 4\cdot(-2) & 1\cdot 2 + 1\cdot 1 + 4\cdot 4 & 1\cdot 0 + 1\cdot 4 + 4\cdot 2 \\ (-2)\cdot(-3) + 0\cdot 1 + 2\cdot(-2) & (-2)\cdot 2 + 0\cdot 1 + 2\cdot 0 & (-2)\cdot 0 + 0\cdot 4 + 2\cdot 2 \end{pmatrix} + \begin{pmatrix} -8 & -12 & 1 \\ -23 & 4 & -11 \\ -16 & -8 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 11 & 4 & 8 \\ 10 & 19 & 12 \\ 2 & -4 & 4 \end{pmatrix} + \begin{pmatrix} -8 & -12 & 1 \\ -23 & 4 & -11 \\ -16 & -8 & 4 \end{pmatrix}$$

$$\therefore D^2 + DE = \begin{pmatrix} 3 & -8 & 9 \\ -13 & 23 & 1 \\ -14 & -12 & 8 \end{pmatrix}$$

f) $C^t \cdot A$

$$\begin{pmatrix} -2 & 7 \\ 3 & -3 \\ -7 & -2 \end{pmatrix}_{3\times 2} ; \begin{pmatrix} 1 & 0 \\ 2 & 7 \end{pmatrix}_{2\times 2}$$

$$\begin{pmatrix} (-2)\cdot 1 + 7\cdot 2 & -2\cdot 0 \; 7\cdot 7 \\ 3\cdot 1 + (-3)\cdot 2 & 3\cdot 0 + (-3)\cdot 7 \\ -7\cdot 1 + (-2)\cdot 2 & -7\cdot 0 + (-2)\cdot 7 \end{pmatrix}$$

$$\therefore C^t \cdot A = \begin{pmatrix} 12 & 49 \\ -3 & -21 \\ -11 & -14 \end{pmatrix}$$

g) E-AC $\begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix} - \begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -2 & 3 & -7 \\ 7 & -3 & -2 \end{pmatrix}$

$\begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix} - \begin{pmatrix} 0\cdot(-2)+5\cdot7 & 0\cdot3+5\cdot(-3) & 0\cdot(-7)+5\cdot(-2) \\ 3\cdot(-2)+7(-2) & 3\cdot3+(-2)\cdot(-3) & 3\cdot(-7)+(-2)\cdot(-2) \end{pmatrix}$

$\begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix} - \begin{pmatrix} 35 & -15 & -10 \\ -20 & 15 & -17 \end{pmatrix}$

$\therefore$ E - A C é indefinido

h) $F^t \cdot E$ $\begin{pmatrix} 1 & -2 & 0 \end{pmatrix} \cdot \begin{pmatrix} 2 & 4 & -3 \\ -1 & 0 & -4 \\ -6 & 0 & -1 \end{pmatrix}$

$\begin{pmatrix} 1\cdot2+(-2)\cdot(-1)+0\cdot(-6) & 1\cdot4+(-2)\cdot0+0\cdot0 & 1\cdot(-3)+(-2)\cdot(-4)+0\cdot(-1) \end{pmatrix}$

$\therefore F^t \cdot E$ $\begin{pmatrix} 4 & 4 & 5 \end{pmatrix}$

i) BCF $\begin{pmatrix} 0 & 5 \\ 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -2 & 3 & -7 \\ 7 & -3 & -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 0\cdot(-2)+5\cdot7 & 0\cdot3+5\cdot(-3) & 0\cdot(-7)+5(-2) \\ 3\cdot(-2)+(-2)\cdot7 & 3\cdot3+(-2)(-3) & 3\cdot(-7)+(-2)(-2) \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 35 & -15 & -10 \\ -20 & 15 & -25 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 35\cdot1+(-15)\cdot(-2)+(10)\cdot0 \\ -20\cdot1+15\cdot(-2)+(-25)\cdot0 \end{pmatrix} \Rightarrow \begin{pmatrix} 65 \\ -50 \end{pmatrix}$

2- a) $C_{2\times4}$ Definido

b) $C_{4\times2}$ Definido

c) Não Definido $2 \neq 3$

d) $C_{5\times3}$ Definido

e) Não Definido $4 \neq 3$

f) $C_{4\times4}$ Definido

g) $C_{2\times3}$ Definido

h) $C_{2\times2}$ Definido

3- a) $A = (a_{ij})_{2\times3}$, onde $a_{ij} = 3i - 2j$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \Rightarrow \begin{pmatrix} 3\cdot1-2\cdot1 & 3\cdot1-2\cdot2 & 3\cdot1-2\cdot3 \\ 3\cdot2-2\cdot1 & 3\cdot2-2\cdot2 & 3\cdot2-2\cdot3 \end{pmatrix}$$

$$\therefore A = \begin{pmatrix} 1 & -1 & -3 \\ 4 & 2 & 0 \end{pmatrix}$$

b) $B = (b_{ij})_{3\times3}$, onde $b_{ij} \begin{cases} 3i+j \text{ ; se } i=j \\ i^2-j, \text{ se } i \neq j \end{cases}$

$$\begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \Rightarrow \begin{pmatrix} 3\cdot1+1 & 1^2-2 & 1^2-3 \\ 2^2-1 & 3\cdot2+2 & 2^2-3 \\ 3^2-1 & 3^2-2 & 3\cdot3+3 \end{pmatrix} = \begin{pmatrix} 4 & -1 & -2 \\ 3 & 8 & 1 \\ 8 & 7 & 12 \end{pmatrix}$$

c) $C = (c_{ij})_{1\times4}$, onde $c_{ij} = j^i$

$$\begin{pmatrix} c_{11} & c_{12} & c_{13} & c_{14} \end{pmatrix} \Rightarrow \begin{pmatrix} 1^1 & 2^1 & 3^1 & 4^1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix}$$

d) $D = (d_{ij})$, onde $d_{ij} = \begin{cases} i^2+j^2, \text{ se } i=j \\ 2ij, \text{ se } i \neq j \end{cases}$

$$\begin{pmatrix} d_{11} & d_{12} & d_{13} & d_{14} \\ d_{21} & d_{22} & d_{23} & d_{24} \\ d_{31} & d_{32} & d_{33} & d_{34} \\ d_{41} & d_{42} & d_{43} & d_{44} \end{pmatrix} \Rightarrow \begin{pmatrix} 1^2+1^2 & 2\cdot1\cdot2 & 2\cdot1\cdot3 & 2\cdot1\cdot4 \\ 2\cdot2\cdot1 & 2^2+2^2 & 2\cdot2\cdot3 & 2\cdot2\cdot4 \\ 2\cdot3\cdot1 & 2\cdot3\cdot2 & 3^2+3^2 & 2\cdot3\cdot4 \\ 2\cdot4\cdot1 & 2\cdot4\cdot2 & 2\cdot4\cdot3 & 4^2+4^2 \end{pmatrix}$$

$$\therefore D = \begin{pmatrix} 1 & 4 & 6 & 8 \\ 4 & 8 & 12 & 16 \\ 6 & 12 & 18 & 24 \\ 8 & 16 & 24 & 32 \end{pmatrix}$$

4. a) Sabendo que $[BA]_{23} = a_{13} \cdot b_{21} + a_{23} \cdot b_{22} + a_{33} \cdot b_{23}$

$\therefore [BA]_{23} = 1 \cdot 2 + 2 \cdot (-1) + 5 \cdot 4$

$\therefore [BA]_{23} = 20$

b) Sabendo que $[AB]_{23} = b_{13} \cdot a_{21} + b_{23} \cdot a_{22} + b_{33} \cdot a_{23}$

$[AB]_{23} = (-2) \cdot 3 + (-3) \cdot 4 + 2 \cdot (-7)$

$\therefore [AB]_{23} = -32$

c) Sabendo que $[B^2]_{31} = b_{31} \cdot b_{11} + b_{32} \cdot b_{21} + b_{33} \cdot b_{31}$

$[B^2]_{31} = -3 \cdot 1 + (-1) \cdot 2 + (-7) \cdot (-3)$

$[B^2]_{31} = -22$

d) $tr(A) = a_{11} + a_{22} + a_{33}$

$tr(A) = 1 + (-3) + 5$

$tr(A) = 3$

e) $tr(B^t) = [B^t]_{11} + [B^t{}_{22}] + [B^t]_{33}$

Sabendo que $tr(B^t) = tr(B) \Rightarrow tr(B) = B_{11} + B_{22} + B_{33}$

$tr(B) = 1 + (-1) + (-7)$

$tr(B) = -7$

f) $tr(A-B) = (A-B)_{11} + (A-B)_{22} + (A-B)_{33}$

Sabendo que $tr(A-B) = tr(A) - tr(B)$, então

$tr(A-B) = 3 - (-7)$

$tr(A-B) = 10$

g) $tr(AB) = (AB)_{11} + (AB)_{22} + (AB)_{33}$

$tr(AB) = 4 + 9 + (-42)$

$tr(AB) = -29$

5- a) $2X + A = 3B + C$

$$2X + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = 3 \cdot \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

$$2X + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 12 & 9 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

$$2X = \begin{pmatrix} 6 & 5 \\ 13 & 9 \end{pmatrix} - \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix}$$

$$X = \begin{pmatrix} 7 & -2 \\ 11 & 3 \end{pmatrix} \div 2$$

$$X = \begin{pmatrix} 7/2 & -1 \\ 11/2 & 3/2 \end{pmatrix}$$

b) $Y + A = \frac{1}{2} \cdot (B - C)^t$

$$Y + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \frac{1}{2} \left\{ \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} - \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \right\}^t$$

$$Y + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \frac{1}{2} \left\{ \begin{pmatrix} 2 & -1 \\ 3 & 3 \end{pmatrix} \right\}^t$$

$$Y + \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 3 \\ -1 & 3 \end{pmatrix}^t$$

$$Y = \begin{pmatrix} 1 & 3/2 \\ -1/2 & 3/2 \end{pmatrix} - \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix}$$

$$Y = \begin{pmatrix} 2 & \frac{11}{2} \\ -5/2 & 9/2 \end{pmatrix}$$

c) $3X + A = B - X$

$4X = B - A$

$X = (B - A) \div 4$

$$X = \left\{ \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} - \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} \right\} \div 4$$

$$X = \begin{pmatrix} 3 & -6 \\ 2 & -3 \end{pmatrix} \div 4$$

$$X = \begin{pmatrix} 3/4 & -3/2 \\ 1/2 & -3/4 \end{pmatrix}$$

D) $\begin{cases} X + Y = 3A \\ X - Y = 2B + C \end{cases}$ +

$2X + 0Y = 3A + 2B + C$

$2X = 3 \cdot \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix} + 2 \cdot \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

$2X = \begin{pmatrix} -3 & 21 \\ 6 & 18 \end{pmatrix} + \begin{pmatrix} 4 & 2 \\ 8 & 6 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$

$2X = \begin{pmatrix} 1 & 25 \\ 15 & 24 \end{pmatrix}$

$X = \begin{pmatrix} 1 & 25 \\ 15 & 24 \end{pmatrix} \div 2$

$X = \begin{pmatrix} 1/2 & 25/2 \\ 15/2 & 12 \end{pmatrix}$

para y: temos $X + Y = 3A$

$\begin{pmatrix} 1/2 & 25/2 \\ 15/2 & 12 \end{pmatrix} + Y = 3 \cdot \begin{pmatrix} -1 & 7 \\ 2 & 6 \end{pmatrix}$

$Y = \begin{pmatrix} -3 & 21 \\ 6 & 18 \end{pmatrix} - \begin{pmatrix} 1/2 & 25/2 \\ 15/2 & 12 \end{pmatrix}$

$Y = \begin{pmatrix} -7/2 & -17/2 \\ -3/2 & 6 \end{pmatrix}$

6 - $A^2 = 2 \cdot A$

$\begin{pmatrix} 1 & 1/x \\ x & 1 \end{pmatrix}^2 = 2 \cdot \begin{pmatrix} 1 & 1/x \\ x & 1 \end{pmatrix}$

$\begin{pmatrix} 1 & 1/x \\ x & 1 \end{pmatrix} \begin{pmatrix} 1 & 1/x \\ x & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2/x \\ 2x & 2 \end{pmatrix}$

$\begin{pmatrix} 1 \cdot 1 + \frac{1}{x} \cdot x & \frac{1}{x} \cdot 1 + 1 \cdot \frac{1}{x} \\ x \cdot 1 + 1 \cdot x & x \cdot \frac{1}{x} + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 2/x \\ 2x & 2 \end{pmatrix}$

$\begin{pmatrix} 2 & 2/x \\ 2x & 2 \end{pmatrix} = \begin{pmatrix} 2 & 2/x \\ 2x & 2 \end{pmatrix}$

$A^n = n \cdot A$

7- a) $A(B+C)$, Sabendo que:

$A(B+C) = A \cdot B + A \cdot C$

$\therefore A(B+C) = X \cdot Y$

b) $B^t A^t$, sabendo que:

$B^t A^t = (A \cdot B)^t$

$\therefore B^t A^t = (X)^t$

c) $C^t A^t$, sabendo que:

$C^t \cdot A^t = (A \cdot C)^t$

$\therefore C^t \cdot A^t = (Y)^t$

d) $(ABA) \cdot C$; Sabendo que

$(ABA)C = AB \cdot AC$

$\therefore (ABA)C = X \cdot Y$

8- a) $A^t = A$

$$\begin{pmatrix} 4 & 2x-3 \\ x+2 & x+1 \end{pmatrix} = \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$$; Sabendo que $a_{ij} = a_{ij}$

$2x - 3 = x + 2$

$x = 5$

tendo $x = 5$:

$$A = \begin{pmatrix} 4 & 7 \\ 7 & 4 \end{pmatrix}$$

b) $B^t = -B$

$$\begin{pmatrix} 0 & x & y \\ -4 & 0 & 2z \\ 2 & 1-z & 0 \end{pmatrix} = \begin{pmatrix} 0 & 4 & -2 \\ -x & 0 & -1+z \\ -y & -2z & 0 \end{pmatrix}$$, Sabendo que $b_{ij} = b_{ij}$

$x = 4$ $\qquad 2z = (-1+z)$

$y = -2$ $\qquad z = -1$

$\therefore x = 4$; $y = -2$; $z = -1$

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$$3\cdot\begin{pmatrix} x & y \\ z & t \end{pmatrix} = \begin{pmatrix} x & 6 \\ -1 & 2t \end{pmatrix} + \begin{pmatrix} 4 & x+y \\ z+t & 3 \end{pmatrix}$$

$$\begin{pmatrix} 3x & 3y \\ 3z & 3t \end{pmatrix} = \begin{pmatrix} x+4 & 6+(x+y) \\ -1+(z+t) & 2t+3 \end{pmatrix}$$

assim, temos:

$3x = x+4$; $2x = 4$; $x = 2$

$3y = 6+x+y$; $2y = 6+x$; $2y = 6+2$; $2y = 8$; $y = 4$

$3z = (-1)+z+t$; $2z = (-1)+3$; $2z = 2$; $z = 1$

$3t = 2t+3$; $t = 3$

$\therefore x = 2 \ ; \ y = 4 \ ; \ z = 1 \ ; \ t = 3$

10-a) $R(\theta)$ é ortogonal se $R(\theta)\cdot(R(\theta))^t = I_n$

$$\begin{pmatrix} \cos\theta & \operatorname{sen}\theta \\ -\operatorname{sen}\theta & \cos\theta \end{pmatrix}\cdot\begin{pmatrix} \cos\theta & -\operatorname{sen}\theta \\ \operatorname{sen}\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} \cos\theta\cdot\cos\theta+\operatorname{sen}\theta\cdot\operatorname{sen}\theta & \cos\theta\cdot(-\operatorname{sen}\theta)+\operatorname{sen}\theta\cdot\cos\theta \\ -\operatorname{sen}\theta\cdot\cos\theta+\cos\theta\cdot\operatorname{sen}\theta & -\operatorname{sen}\theta\cdot(-\operatorname{sen}\theta)+\cos\theta\cdot\cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} \cos^2\theta+\operatorname{sen}^2\theta & 0 \\ 0 & \operatorname{sen}(\theta)^2+\cos(\theta)^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$ $\therefore$ a matriz $R(\theta)$ é ortogonal

b)
$$\begin{pmatrix} 1 & 0 & x \\ 0 & \frac{1}{\sqrt{2}} & y \\ 0 & \frac{1}{\sqrt{2}} & z \end{pmatrix}\cdot\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ x & y & z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1\cdot1+0\cdot0+x\cdot x & 1\cdot0+0\cdot\frac{1}{\sqrt{2}}+x\cdot y & 1\cdot0+0\cdot\frac{1}{\sqrt{2}}+x\cdot z \\ 0\cdot1+\frac{1}{\sqrt{2}}\cdot0+y\cdot x & 0\cdot0+\frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}+y\cdot y & 0\cdot0+\frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}+y\cdot z \\ 0\cdot1+\frac{1}{\sqrt{2}}\cdot0+z\cdot x & 0\cdot0+\frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}+z\cdot y & 0\cdot0+\frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}+z\cdot z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1+x^2 & xy & xz \\ xy & \frac{1}{2}+y^2 & \frac{1}{2}+yz \\ xz & \frac{1}{2}+yz & \frac{1}{2}+z^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$1 + x^2 = 1$
$x^2 = 0$
$x = 0$

$\frac{1}{2} + y^2 = 1$
$1 + 2y^2 = 2$
$2y^2 = 1$
$y^2 = \frac{1}{2}$
$y = \pm \frac{1}{\sqrt{2}}$
$y = \pm \frac{\sqrt{2}}{2}$

$\frac{1}{2} + z^2 = 1$
$z = \pm \frac{1}{\sqrt{2}}$
$z = \pm \frac{\sqrt{2}}{2}$

$\frac{1}{2} + y \cdot z = 0$

Se $y = +\frac{\sqrt{2}}{2}$, então
$\frac{1}{2} + \frac{\sqrt{2}}{2} \cdot z = 0$
$1 + \sqrt{2} \cdot z = 0$
$\sqrt{2} \cdot z = -1$
$z = -\frac{1}{\sqrt{2}} \Rightarrow -\frac{\sqrt{2}}{2}$

ou se $y = -\frac{\sqrt{2}}{2}$, então
$z = +\frac{\sqrt{2}}{2}$